INFINITY RACING
ESTD. 2022

aramco
F1 in Schools
SINGAPORE | 2023
WORLD FINALS

PROJECT MANAGEMENT
PORTFOLIO



LUCKY
ENCOURAGING INNOVATION

RAYO RACING
INFINITY Premium Motoring CARS

Ansys
HOP HOUSE OF POLISH

KeyShot
FORGE3D

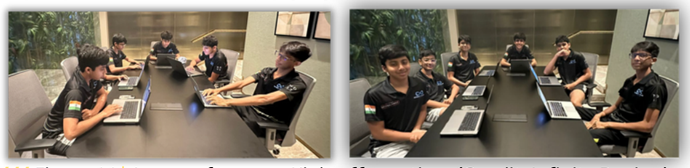
ABEC 3 5 7

1.1 /// KICK-OFF **MEETING**

The Kick-Off Meeting was conducted for the continued success of our team after our Nationals campaign and to maintain our team morale as we embark on a new journey to the World Finals.

The Kick-Off Meeting played a crucial role as we ventured onto a new project. In this meeting, we brainstormed pivotal factors on how to move ahead as we progressed to the World Finals, such as constraints, risks, roles and responsibilities. We reflected upon our past projects and scrutinised where we could improve in terms of key project deliverables. Although we kicked-off a new project, we re-established our primary goal of becoming World Champions.

***KICK-OFF MEETING DATE:* 01/12/2022**



/// *Figure 1.1 | Images from our Kick-off meeting. (Credit: Infinity Racing)*

1.2 /// FINAL **SCOPE STATEMENT**

Our primary is to be crowned World Champions. In order to achieve this goal, we created two divisions, an In scope and an Out Of Scope section. In Scope consists of all the deliverables that must be done, while Out Of Scope consists of means elements that are outside the parameters of the project. This was critical to initialise the project as it directed us towards the path we needed to take.

IN SCOPE	OUT OF SCOPE
Car abiding by set regulations	Avoid plagiarism at all costs
Portfolios adhering by set regulations	Use of materials that are not sustainable
Verbal Presentation within 10-minute limit	Self-funding due to insufficient acquirement of sponsorship funds
Pit Display abiding with set regulations	Preparing for participating in future competitions.
Quality Acceptance Criteria should be met for all deliverables	Deliverables not abiding by set regulations

1.3 /// PROJECT **GOALS**

Right from the beginning of our project, we set certain goals to be aimed to complete by the end. Our primary goal was set: To be crowned F1 in Schools 2023 World Champions.
The standard goals we set were:

- Car complying with all technical regulations.
- Portfolios following scorecard and improvement from National Finals based on feedback from mentors.
- Achieve victory in the Project Management award and Pit Display award.
- All project elements were submitted on time.

1.4 /// PROJECT **CHARTER**

A project charter was developed to authorise our project. In order to do this, we described our project purpose, goals, deliverables, stakeholders, constraints, assumptions, milestones and scope statement.

The project charter was the first step to initialise our World Finals project and to begin our journey as a team.

PROJECT TITLE	F1 IN SCHOOLS WORLD FINALS 2023	**DATE STARTED**	1/12/2022 (1st December 2022)
PROJECT MANAGER	LAKSHYA HIRAWAT		

PROJECT DESCRIPTION

For this project, our primary goal was to be World Champions. In order to achieve this goal, we are required to complete all project deliverables while submitting them on time. This included 3 portfolios, 2 identical air-powered cars, a 10-minute Verbal Presentation, A Pit Display incorporating our team identity and values, and Annexures.
In order to achieve our primary goal, we had a set of objectives that we followed, such as:
Developing a car that complies with all technical regulations while maintaining a high level of performance. Improvising on all our portfolios from our previous projects and lastly presenting an aesthetic as well as an informative pit display.

TEAM MEMBERS	RESPONSIBILITIES
Lakshya Hirawat	Team Manager, Graphic Designer
Aryaman Kejriwal	Team Manager, Design Engineer
Nathan Doshi	Sponsorship Manager, Manufacturing Engineer
Raghav Sharma	Graphic Designer, Sponsorship Manager
Rajvir Sehgal	Graphic Designer, Resource Manager
Yuvraj Ruparel	Design Engineer, Manufacturing Engineer

KEY STAKEHOLDERS

Team Members	Sponsors & Partners	F1 In Schools

STAKEHOLDER ENGAGEMENT

STAKEHOLDER	CONTACT
STEMplify India	f1s@timeofsports.com
Lucky Investments	+91 98204 70695
KeyShot	education@luxion.com
Ansys	vishal.ganore@ansys.com
Corugami	udit@corugami.com
ABEC357	info@abec357.com
House of Polish	info@abec357.com
Forge 3D	contact.forge3d@gmail.com

CONSTRAINTS

Track Testing: For F1 In Schools India, track testing was only available in Delhi. However, our team was based in Mumbai making track testing limited for our team.

Sponsor Search: The team relies on sponsorships for funding of the entire campaign. Limited sponsors could lead to a negative impact on our performance and quality of work.

Limited time rendering: Our team received a 2 month licence for rendering on Luxion Keyshot which was only a limited time frame for rendering.

Bearings imported: Bearings equipped on our car were imported from the USA which took a long time to reach us leading to a time constraint.

ASSUMPTIONS

World Finals held physically
Increase of costs due to travel and stay at hotel
All project deliverables in offline format
Physical Pit Display and Verbal Presentation
Current sponsors will continue to support us

MAJOR MILESTONES

Completing design of car	Manufacturing & Finishing of car is complete	All Portfolios complete	All Project deliverables submitted
20/05/2023	25/06/2023	15/08/2023	1/09/2023

RESOURCES

Entire team
Workplace (Beaumonde Towers)
Stakeholders (Team, Sponsors, F1 in Schools)
Communication Platforms (WhatsApp, Email)
Printing Agency (Parksons Graphics)
Engineering Softwares (Fusion 360, Ansys, KeyShot)
Graphic Designing Softwares (Canva, Adobe Photoshop)
3D Printing Services (Forge 3D)

PROJECT DELIVERABLES

ENGINEERING
2x Identical Cars (Car A and B)
1x F1 Model Block
11 - page Design & Engineering Portfolio
Replacement Components
A4 Engineering Drawings
A4 Car Renders
Official F1 Model Block holographic sticker

ENTERPRISE
11 - page Enterprise Portfolio
7 - page Project Management Portfolio
Physical Pit Display
10 - minute Verbal Presentation
'Team Partnerships' Declaration(s) Documents
Project Elements Submission Checklist

1.5 /// QUALITY **ACCEPTANCE CRITERIA**

CATEGORY	QUALITY ACCEPTANCE CRITERIA	ASSESSMENT METHOD	TIMING TO REVIEW	RESPONSIBILITY	APPROVAL
2x Identical Cars	All internal and external dimensions must comply with set Technical Regulations.	Inspecting Orthographic Projections for Internal Dimensions. Measuring components using Vernier Calliper for External Dimensions.	Before and After Manufacturing and Finishing of both cars.	Engineering Team	Team Manager & Design Mentors
	F1 in Schools decal must be visible at all times.	Visual Inspection	After pasting decals on the car.	Manufacturing Engineer	Team Manager
	No parts should suffer from breakage or damage of any kind	Physical testing on each component including Stress Test, Drop Test and post-test evaluation.	Before and After Manufacturing and Finishing of both cars.	Manufacturing Engineer	Team Manager
	The car must withstand forces experienced during racing	Physical testing including Track Testing.	After Manufacturing and Finishing of both cars.	Manufacturing Engineer	Team Manager
Portfolios	All documents must comply with set scorecard regulations in Competition Regulations	Visual Inspection	Before printing of all portfolios	Project Manager	Enterprise Mentors
	All Content must be approved by mentors.	Checklist used to ensure all pages of portfolios have been assessed and approved by mentors.	Before printing of all portfolios	Team Manager	Enterprise Mentors
Pit Display	Display dimensions must comply with set Competition Regulations.	All components of Pit Display measured using measuring tape.	Before and after manufacturing and finishing of pit display.	Design Engineer	Team Manager & Design Mentors
	Display must effectively showcase team's brand identity	Visual inspection to ensure team logo, team name and team colours are incorporated in the Pit Display.	Before manufacturing of Pit Display	Graphic Designer	Team Manager
	Display must be made of durable materials	Multiple tests as well as research conducted on the material used.	Before manufacturing of Pit Display	Resource Manager	Resource Manager & Design Mentors
Verbal Presentation	Presentation must not exceed set time limit of 10 minutes	Prepared multiple specimen presentations which were timed using a stopwatch.	Prior to finalising all Presentation dialogues and presentation slides.	Enterprise sub-team	Team Manager
	Presentation must be approved by mentors.	Checklist used to ensure all slides of presentation have been assessed and approved by mentors.	Prior to completing the final draft for Verbal Presentation.	Enterprise sub-team	Enterprise Mentors

EVALUATION

A kick of meeting was conducted, discussing various critical aspects before the start of the project. The project charter was created to authorise the project, aiding in developing the scope statement and project goals. A Quality Acceptance Criteria was critical for each deliverable's success.

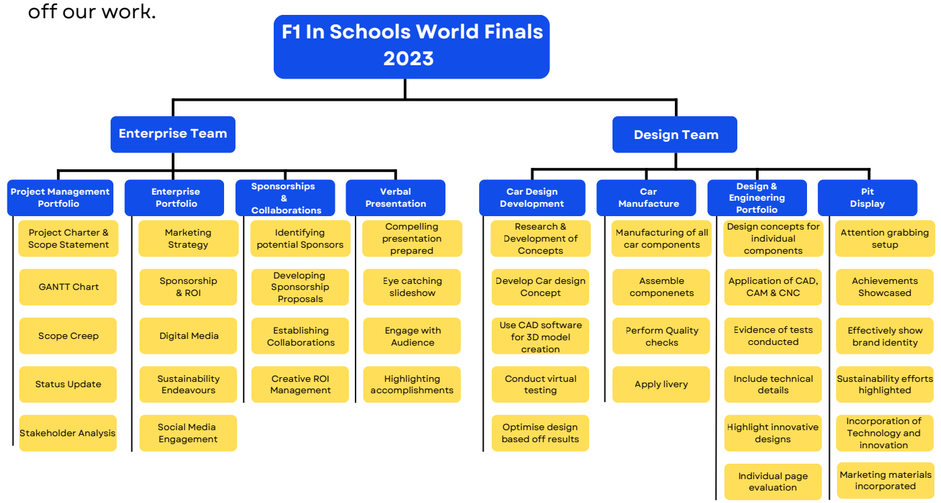
PROJECT SCHEDULE

INFINITY RACING

2.1 /// WORK **BREAKDOWN STRUCTURE**

A work breakdown structure was one of our initial processes and outlined all of the tasks that we were required to complete such as all project deliverables. It provided us with a clear understanding of how much work each team member would be loaded with during the project and how the major category of work has been broken down to the entire team.

Our project was broken down to all the project deliverables which were further broken down into smaller components to make our work easier and manageable in terms of workload. The Work Breakdown Structure provided us with an elaborate plan for managing and organising our work and acted as an initial pillar to our project and to kick off our work.



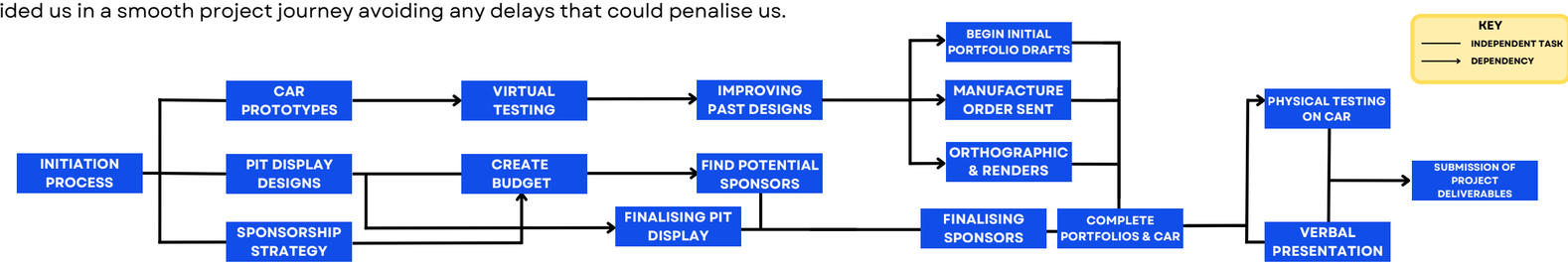
PROJECT SCHEDULE

The project schedule was crucial for our team as it outlined the path we would take through our working period. It defined the workload on each member of our team respective of their roles and responsibilities in the team.

The development of the Project Schedule aided us in creating our preliminary project timeline and the GANTT chart for identifying specific time durations and dependences on project deliverables.

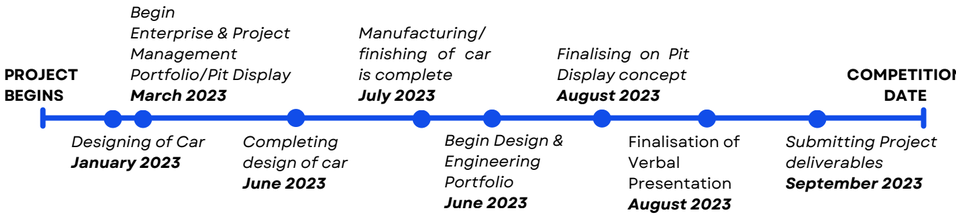
2.2 /// CRITICAL **PATH METHOD**

The Critical Path method associates the crucial work that has to be accomplished during a set time period through the duration of the project. Analysing our Project Schedule through this method helped us identify the time period and dependencies of all deliverables in the project. We incorporated our major category of work in the form of the Critical Path to avoid time delay and identify needed dependencies through our project. Our path consisted of 7 primary stages starting with initialising our project and ending with the completion and submission of all the project deliverables. The use of the Critical Path method aided us in a smooth project journey avoiding any delays that could penalise us.

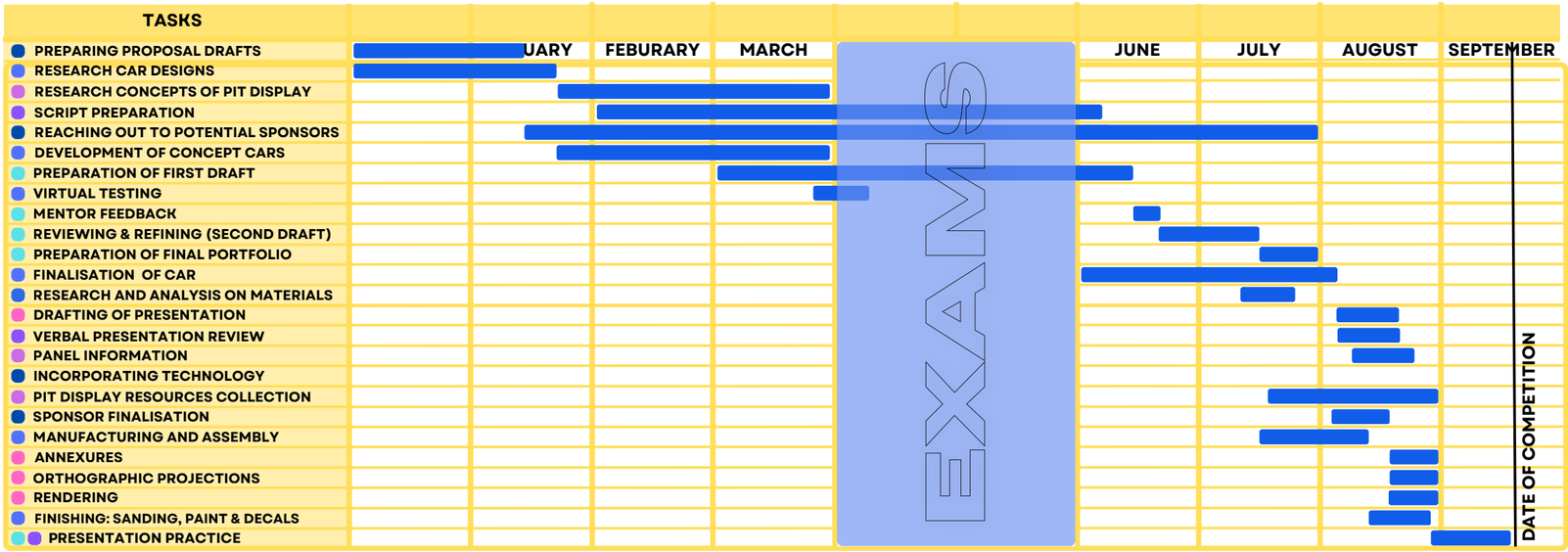


2.3 /// PROJECT **TIMELINE**

To avoid our time constraint and any penalisation for time delay, we developed a preliminary project schedule highlighting dates of important events through the duration of the project till the competition date. The project timeline mainly marks major milestones and important events with their respective dates.



2.4 /// GANTT **CHART**



3.1 /// BUDGET ESTIMATION

Since the World Finals was a large project, estimating a budget was essential for our team. We used basic budgeting techniques to estimate a budget for the project. We used a Range-based budgeting approach and substituted a formula in order to estimate the budget accurately.

We estimated a maximum and a minimum budget in case of deviation and the possibility of risks. A Base Budget was developed as the desired budget for the entire project. The Maximum/Minimum percentage represents the range/percentage of deviation that could occur from the Base Budget. Hence, the following formula was used:

Maximum Budget = Base Budget + (Base Budget × Maximum Percentage)
Minimum Budget = Base Budget - (Base Budget × Minimum Percentage)

According to previous competitors and our previous project, we estimated our Base Budget to be $24,200 (USD). Using the possibility and impact of each risk, we calculated our Maximum Percentage as 6% and Minimum Percentage as 2.5%. Therefore, we calculate our estimated budget as follows:

Baseline Budget = $24,200
Maximum Percentage = 8%
Minimum Percentage = 5.5%

Maximum Budget = 24,200 + (24,200 × 0.08)
= $26,136

Minimum Budget = 24,200 - (24,200 × 0.035)
= $22,869

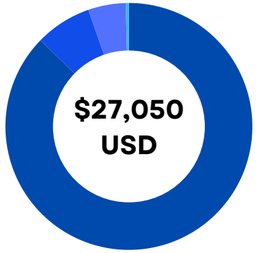
Therefore, we calculated our estimated budget as shown above.

3.2 /// BUDGET CONTROL

In order to formulate an accurate budget, we used various budget-controlling techniques which greatly aided us. These methods are frequently used by various organisations and have proven time and again to be efficient methods of budgeting.

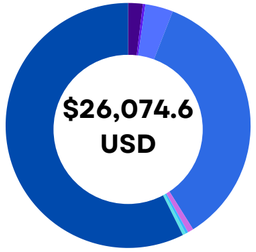
- *Value-Based* Budgeting: This budgeting method was utilised to efficiently allocate our funds to areas that would be of great value towards our project. This generally meant allocating necessary funds to project deliverables.
- *Benchmarking:* We approached past Indian World Finals teams who attended the physical event, requesting an approximation of their budget. This was then analysed and compared to our formulated budget and helped us identify areas where we could control and cut down our expenses.
- *Budgetary Slack Management:* Initially, we deliberately overestimated our total costs and underestimated the amount we were going to raise from sponsors. This method allowed for better performance evaluation in dynamic environments.

INCOME BUDGET



SOURCE OF INCOME	ESTIMATE	ACTUAL
Sponsors	$20,500	$23,450
Donations	$1250	$1900
Fundraising	$1200	$1520
Events & Merchandise	$100	$180

EXPENDITURE BUDGET



SOURCE OF INCOME	ESTIMATE	ACTUAL
Car Manufacturing	$600	$467
Car Finishing	$120	$91
Bearings	$1500	$994
Event Registration	$8500	$9650.58
Pit Display	$300	$201
Portfolios	$120	$91
Team Apparel	$180	$108
Travel	$17100	$14472

3.3 /// RESOURCE MANAGEMENT

Managing the resources we utilised throughout the project was crucial for the team. We registered each resource we used by mentioning the date of requirement (when the resource was required), the resource provider (where we acquired our resources from), how prevention of resource levelling was executed and how necessary actions were performed in case of resource smoothing. This aided in allocating each of the resources that were utilised and prevented any constraints to act upon the team. Resource Managing also enabled us to navigate through how much of our budget was allocated to each resource utilised.

EVIDENCE OF RESOURCES USED:







RESOURCE REQUIRED	WHEN (DATE OF REQUIREMENT)	RESOURCE PROVIDER	HOW RESOURCES WERE UTILISED	ACTION INCASE OF RESOURCE SMOOTHING
3D Printed Car Components	03/07/2023	Forge 3D	Assembly of 2x identical cars	Recheck car dimensions sent to ensure smooth printing
Bearings	09/07/2023	ABEC357	Assembly of final car's wheel system	Reduce use or reuse previous bearings for track testing
Model Block	25/06/2023	Denford Ltd.	Assembly of 2x identical cars	Avoid any errors in CNC machining
Car Livery Application	22/07/2023	Multiple sources	Finishing of 2x identical cars	Keep complexity of livery standardised to avoid errors
Decals	23/07/2023	Multiple sources	Finishing of 2x identical cars	Preplan decal placement to avoid error in pasting
CO2 Canisters	02/08/2023	Denford Ltd.	Track testing of final and prototype cars	Reduce excess use and avoid false tasts during testing
Tetherline	02/08/2023	Denford Ltd.	Track testing of final and prototype cars	Reuse previously used tetherlines
Racetrack	02/08/2023	STEMplify	Track testing of final and prototype cars	Reuse older race tracks.
Wind Tunnel	23/07/2023	STEMplify	Physical wind tunnel testing of cars	Reuse National Finals wind tunnel
3D Printer	17/06/2023	Forge 3D	Manufacturing of 2x identical cars	Avoid misprint to reduce wastage of printing filament.
Pit Display materials	01/08/2023	Corugami	Assembly of final Pit Display	Priorly test material strength before printing to avoid errors.
Ansys (CAA Software)	18/06/2023	Ansys	Virtual testing of cars	Needed
KeyShot (Rendering Software)	02/07/2023	KeyShot	Submission of A4 Car renders and used in Car reveal	Begin 2 month license when required most
Monetary Sponsorship	-	Multiple Sources	Payment for materials used in project	Search potential sponsors and create attractive proposals.
Team Uniform	10/08/2023	PrintVision	At the World Finals event and other meetings and social media posts	Reuse National Finals uniform

EVALUATION

Estimating a budget prior to beginning was crucial in terms of managing the project. Various budget control techniques were used to get an accurate result for the expected budget and how to minimise it. The resources utilised were listed making it easy to track the team's expenditures.

ROLES & RESPONSIBILITIES

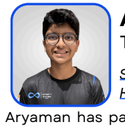
4.1 /// TEAM **MEMBERS**



LAKSHYA HIRAWAT
TEAM MANAGER, GRAPHIC DESIGNER

SKILLS: Canva Pro, Adobe Design Suite, KeyShot, HTML
KNOWLEDGE: Marketing, Aerodynamics, Video Editing, Web Dev, Social Media

Lakshya was the school's FIRST Tech Challenge team captain, sharpening his leadership qualities and enabling him to efficiently manage and handle the team. By knowing a range of software relating to graphic design and participating in numerous designing and technology competitions, Lakshya was fit to be the team's graphic designer.



ARYAMAN KEJRIWAL
TEAM MANAGER, DESIGN ENGINEER

SKILLS: Fusion 360, Solidworks, Ansys, Python, Excel, QuickCam Pro
KNOWLEDGE: Aerodynamics, Material Properties, CAA, CAD/CAM

Aryaman has participated in the FIRST Tech Challenge for two years and also participated in the FIRST Robotics Challenge 2023, giving him an edge in mastering CAD and CAM applications. Aryaman has also accomplished the Les Elfes Leadership Course, enabling him to organise and lead the team.



NATHAN DOSHI
SPONSORSHIP MANAGER, MANUFACTURING ENGINEER

SKILLS: Adobe Design Suite, QuickCam Pro, Mailchimp, Trello, Excel, LinkedIn
KNOWLEDGE: Aerodynamics, Financial Management, Sponsorship Management

Nathan is a skilled writer who secured the 1st position in the IEO Olympiad hosted by SOF. His excellent writing skills and persuasive language make him perfect for the role of Sponsorship Manager. His knowledge of CAM applications like QuickCam Pro and manufacturing makes him suitable for the team's Manufacturing Engineer.



YUVRAJ RUPAREL
DESIGN ENGINEER, MANUFACTURING ENGINEER

SKILLS: Fusion 360, Solidworks, KeyShot, Ansys
KNOWLEDGE: Aerodynamics, Kinematics, Material Properties, CAA, CAD/CAM

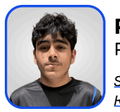
Yuvraj has participated in the FIRST Robotics Challenge, making him efficient in many engineering software and suitable as a design and manufacturing engineer. He has also attended the John Hopkins Centre of Talented Youth where he learned engineering and Arduino Prototyping.



RAGHAV SHARMA
GRAPHIC DESIGNER, SPONSORSHIP MANAGER

SKILLS: Canva Pro, CSS, Adobe Design Suite, KeyShot, Ansys, Filmora, LinkedIn
KNOWLEDGE: Rendering, CAA, Web Dev, Video Editing

Raghav has successfully completed a course on Sustainability and CAD from the prestigious Brown University, making him fit for the competition. He has also completed the 20-week Young Entrepreneur's Academy, giving him the essential learnings of handling finances and making him qualified for the Sponsorship Manager role.



RAJVIR SEHGAL
RESOURCE MANAGER, GRAPHIC DESIGNER

SKILLS: Canva Pro, Adobe Design Suite, Excel, KeyShot
KNOWLEDGE: Rendering, Material properties, Meta Business, Image analytics

Rajvir has primarily worked on the content and design of Pit Display. He majorly assisted in the slideshow and content of the Verbal Presentation. Researched valid materials that can be used for the car and Pit Display and helped obtain resources for the entire project. Also, he aided in the resource and budget management of the project.

4.2 /// JOB **FUNCTIONS**

TEAM MANAGER
- Oversees the overall functioning of the team
- Responsible for all project deliverables to be submitted
- Ensures effective communication and coordination within the team
- Maintains a high team morale
- Manages team collaborations

Deliverables: Project Management Portfolio, Enterprise Portfolio

DESIGN ENGINEER
- Conducts research on car design
- Develops car prototypes using numerous concepts
- Conducts simulations on the prototypes to analyse and evaluate car designs

Deliverables: Car Design, Design and Engineering Portfolio, Orthographic Projections, Car Renders

MANUFACTURING ENGINEER
- Manages manufacturing process
- Ensures smooth execution of car manufacture
- Handles manufacturing, assembly, and finishing

Deliverables: Car Manufacturing, Design and Engineering Portfolio

GRAPHIC DESIGNER
- Develops visually appealing social media content
- Creates templates and graphics for team deliverables
- Designs appealing Pit Display making it aesthetic for viewers

Deliverables: Car Renders, Portfolio Templates, Pit Display

SPONSORSHIP MANAGER
- Reaches out to potential sponsors
- Maintains relations with past sponsors
- Develops appealing sponsorship prospectus'
- Presents proposals during sponsorship meetings

Deliverables: Project Management Portfolio, Enterprise Portfolio, Car Decal Placement

RESOURCE MANAGER
- Formulates project budget and finances
- Identifies project resources
- Selects service/ product provider for tasks

Deliverables: Project Management Portfolio, Enterprise Portfolio, Design and Engineering Portfolio, Pit Display

4.3 /// TEAM **STRUCTURE**

A highly detailed structure is pivotal to creating a well-functioning team that can smoothly conduct all operations required in the project. All role duties and responsibilities were clearly mentioned for all designations that were there.

The Work Breakdown Structure and Critical Path Method were used as important tools to determine our overall team structure for the World Finals.

The team is organised into a two-step hierarchy, with the Team Manager being at the top of the structure. The rest of the team is then organised one rank below, consisting of Design Engineers, Sponsorship Managers, Graphic Designers and other roles.

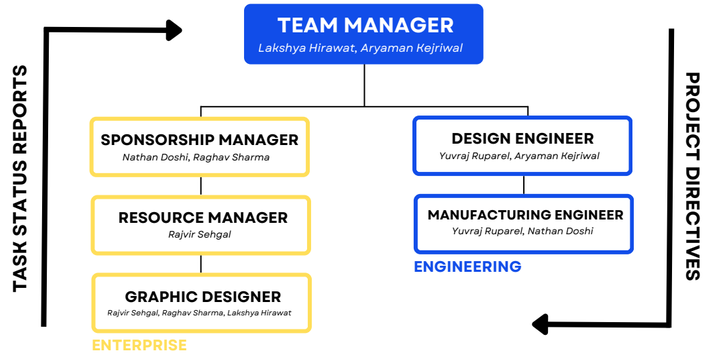
The structure helps everyone to effectively communicate with each other. All decisions are made by the Team Manager, with the rest of the team making sure to give their input on the resolution. Updates are sent about the various project deliverables from the lower rank to the team manager.

4.4 /// RACI **MATRIX**

TASK	TEAM MANAGER	DESIGN ENGINEER	MANUFACTURING ENGINEER	SPONSORSHIP MANAGER	RESOURCE MANAGER	GRAPHIC DESIGNER
Car Design	A	R	C		I	
Car Manufacture	A	I	R		R	
Sponsorship Prospectus	R/I			R	C/A	R
Verbal Presentation	R/C			R		C/I
Pit Display	C	C	R	I	R	R
Project Management Portfolio	C/I			C		A
Enterprise Portfolio	C/I			R		A
Design & Engineering Portfolio	C/I	R	R		C	A
Orthographic Projections	I	R	C			
Car Renders	I	R	C			C
Digital Media	C			I		R
Team Uniform	C/I			A	R/C	R
Project communication	A			R	I	

RESPONSIBLE ACCOUNTABLE CONSULTED INFORMED

TEAM STRUCTURE DIAGRAM



/// *Figure 5.1 | Diagram showing breakdown of team structure. (Credit: Infinity Racing)*

COMMUNICATIONS & RISK MANAGEMENT

INFINITY RACING

5.1 /// COMMUNICATION **PLAN**

Communication is the sending and receiving of information within a project. In order to maintain smooth and efficient communication, we developed a concise Communication Plan for efficient intra-team communication as well as team-stakeholder communication. This was crucial for the project's success since communication and interaction are vital in a project.

The Stakeholder Register and Communication Methods (5.2) were pivotal input tools to create a communication plan.

STAKEHOLDERS	CONTACT	INFORMATION COMMUNICATED	FREQUENCY	METHOD
Team Members	info@infinityracing.in	Project work updates and further tasks	Daily	WhatsApp, Trello
Ansys	vishal.ganore@ansys.com	Software liscense/information	Biweekly	Email
Keyshot	education@luxion.com	Rendering liscense	Monthly	Email
Corugami	udit@corugami.com	Pit display information	Weekly	Email
Abec357	info@abec357.com	No. of bearings/Bearing type	Monthly	Email
Forge 3D	contact.forge3d@gmail.com	Car manufacturing details	Weekly	Email
Lucky Investments	+91 98204 70695	Competition information/Team updates	Weekly	WhatsApp, Meetings
Pallavi Mahar	pallavi@stemplify.com	Designing Mentorship	Weekly	Email, Zoom
Yashraj Singh	yashraj@stemplify.com	Team Updates	Monthly	Email
Shubham Tanjea	shubham@stemplify.com	CAM Mentorship	Monthly	Email, Zoom
House of Polish	+91 99100 85979	Finishing of the Car	Weekly	WhatsApp
WOL 3D	+91 99675 75518	Manufacturing of the Car	Monthly	WhatsApp
Rayo Racing	+91 88797 55705	Team/Competition Updates	Monthly	WhatsApp

For efficient and effective team-to-stakeholder communication, our stakeholders were categorised into the frequency at which they were contacted, the information communicated and the method used. This allowed for easier and quicker communication between the team and the stakeholders.

Our communication plan also included the use of various communication methods which enabled us to communicate within ourselves and with our stakeholders in an organised and professional manner.

5.3 /// RISK **ASSESSMENT**

PROJECT RISKS	LEVEL OF SEVERITY	LEVEL OF POSSIBILITY	IMPACT	OUTCOME	RESPONSE ACTION	AREA AFFECTED
Damage to materials required in Pit Display	Unacceptable	Not Likely	High	Poor aesthetics of Pit Display due to damage	Use of reserve materials for Pit Display carried to venue.	Pit Display
Loss of bearings while travelling	Unacceptable	Not Likely	High	Poor performance of car in racing leading to loss of valuable points	Use of previous bearings being carried to venue as well.	Racing on event
Breakage of car components during travelling	Unacceptable	Possible	High	Poor performance of car in racing.	Quality assessment on all components to ensure no breakages in car	Final physical testing on cars
Car dimensions not complying with regulations	Unacceptable	Possible	High	Poor performance of car in racing leading to added time due to penalties	Follow technical regulations strictly and use quality acceptance criteria for each component.	Car manufacturing and assembly.
Issues/delay with delivery of components	Moderate	Possible	Medium	No excess components during racing or overall delay in manufacturing process	Order parts earlier and keep excess days incase of any possible delay.	Final car manufacturing, finishing and assembly
Scope Creep	Acceptable	Likely	Medium	Delay in Project Schedule and increase in requirement of resources.	Development of an Action Plan and re-allocation of resources.	All possible project deliverables.
Failing to raise adequate amount of funds	Acceptable	Possible	Low	Issues with obtaining valuable resources/Possibility of self-funding	Reach out to potential sponsors or in-kind partners who can help reduce overall costs.	All possible project deliverables.
Failing to acquire any beneficial project collaborations	Moderate	Not Likely	Medium	Higher costs without collaborators/Possibility of self-funding	Find potential collaborators and use close contacts for gaining collaborators.	Enterprise/Project budget
Delay in receiving funds from sponsors	Moderate	Not Likely	Low	Delay in obtaining valuable resources	Stay in touch with sponsors when obtaining funds and constantly remind all sponsors.	All possible project deliverables.
Losing a teammate due to illness, school examinations	Acceptable	Possible	High	Higher workload on specific project deliverables like building of Pit Display.	Early divison of work incase a team member is not available which ensures team is prepared.	Direct impact on project scope

5.2 /// COMMUNICATION **METHODS**

MICROSOFT TEAMS

Frequency: Monthly
Reason: Organisational communication methods to maintain smooth workflow within the team and for interacting with key stakeholders.

GODADDY (EMAIL)

Frequency: Bi-weekly
Reason: Domain based platform for professional emails to potential and existing sponsors. Used for official and important emails within the team as well.

ZOOM

Frequency: Weekly
Reason: Video based method of communication. Interactive features allow intra-team communication as well as communication with stakeholders.

WHATSAPP

Frequency: Daily
Reason: Message based platform primarily used for informal intra team communication and interactions.

TRELLO

Frequency: Bi-weekly
Reason: List based application used for status report and task completion within the team. Used for planning based communication.

MEETINGS

Frequency: Weekly
Reason: Face to face interaction for effective communication between team members and stakeholders.

5.4 /// AREAS OF **IMPACT**

In our project, we managed to recognise the importance of identifying the potential risks and the impact they would have on our team's success. We conducted a thorough analysis to determine exactly which fields would be directly impacted in case of any of these faults, given in the table below. The areas of impact were further categorised into four, them being: Schedule, Resources, Scope and Quality. In this method we could identify and address risks in a much more structured and efficient manner. It allowed us to implement systematic mitigation methods and allocate our resources accordingly to all departments, thus helping to maintain a high standard throughout the course of the project.

PROJECT RISKS	RESOURCES	SCHEDULE	SCOPE	QUALITY
Damage to materials used in Pit Display	■		■	
Damage to car while travelling	■		■	
Breakage Of car components	■		■	■
Car Dimensions Not Complying With Technical Regulation			■	■
Issues/Delay With Delivery Of Components	■	■		

■ AREA OF IMPACT

EVALUATION

A communication plan was critical for easy interaction and collaboration with our key stakeholders. Hence, we developed a communication plan and listed the tools used for communication. We identified potential risks and analysed response actions and areas that would be impacted in the project through Resources, Schedule, Scope and Quality.

6.1 /// STATUS **REPORTS**

It is a document that compromises content which gives an insight into the state that the project is in at the moment. Status reports must be efficient in delivering information while being simple as well. The team members working in their corresponding departments create the initial drafts of the status report. These documents were then sent out to the Team Manager responsible for the respective departments, who would then sign it off and forward it to all of our stakeholders, ranging from sponsors to mentors.

The status report was broken down into - resources utilised, adherence to the project schedule, a possible scope creep or areas of concern and lately a final task sign-off. This standard form of our reports ensured that these documents communicated the information that we had desired to all of our key stakeholders as it is of utmost importance that they must be informed of the project at all times.

STATUS REPORT (31/07/23)

CURRENT TASKS	COMPLETION	ADHERENCE TO SCHEDULE	RESOURCES REQUIRED	AREAS OF CONCERN
Project Management Portfolio			Printing	None
Enterprise Portfolio			Printing	None
Design & Engineering Portfolio			Printing	Final car design should be complete
Final Car Design			CAD/CAM software, Ansys Software	Not on Schedule
Virtual Testing			Ansys license	None
Physical Testing			Track, Canisters, Tetherline, Wind Tunnel	Travel required leading to delay in schedule
Team Uniforms			Apparel Printing	None
Pit Display Layout			CAD Software	None
Pit Display Panel layout			Pit Display materials	None
Final Car Manufacturing			3D Printer, CNC Machine, Model Block, Bearings	Not on Schedule

OVERALL PROJECT HEALTH:

KEY

- COMPLETE/ON SCHEDULE
- IN PROGRESS
- NOT STARTED/OFF SCHEDULE

VIEW OUR PREVIOUS STATUS REPORTS

SCAN

6.2 /// SCOPE **CREEP**

We believed it was crucial to identify scope creep as soon as it arose, and devised several measures to be taken in case of the incident occurring and also various methods to prevent it altogether. The team analysed and created multiple charts regarding resources and various other elements to be clearly defined and made sure that all the requirements were met to get as accurate an estimate as possible.

In case of the unfortunate event of a scope creep, we communicated with stakeholders to determine the course of action to be taken. After much discussion, conclusions would be made on how the situation must be handled henceforth and requested additional resources.

Our team faced a Scope Creep which we identified through the course of the project. We noticed that the design of our final car went over our preliminary developed project schedule and hence was categorised as a Scope Creep. This had a direct impact on our resources and delayed our manufacturing process all the more impacting our Project Schedule. The Scope Creep was resolved before we closed our project. We developed a quality criteria for our identified Scope Creep to make sure there was no compromising in quality due to a scope creep.

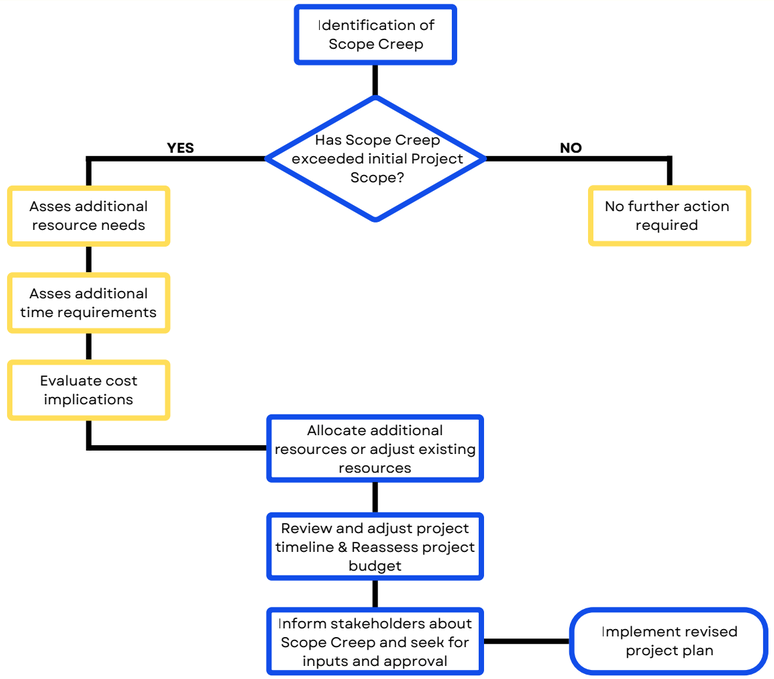
IDENTIFIED SCOPE CREEP	FUTURE TASKS AFFECTED	METHOD TO AVOID QUALITY COMPROMISE
Delay in Designing of car	▸ Manufacturing of final car ▸ Assembly and Finishing of final car ▸ Design and Engineering Portfolio ▸ Rendered Images, Orthographic Projections	▸ Reassess identified scope creep ▸ Adjust project schedule and allocation of resources ▸ Break down scope creep task ▸ Assessment of risks ▸ Reconduct quality criteria assessment once task is complete

/// *Figure 7.1 | Table showing method of not compromising quality even though scope creep is identified (Credit: Infinity Racing)*

ACTION PLAN

A clear Action Plan was developed in case any tasks that overrun the project occurred. This plan was carefully created by our Team Managers and was distributed to each team member in case a task had overrun. The Action Plan was crucial for the entire project to adhere to our planned schedule and to be completed in the required time.

The Action Plan was created as a flowchart to mitigate any overrunning tasks and aided in easier management of tasks. The Action Plan helped in controlling and managing the entire project. Not only this, but the development of an Action Plan aided in risk mitigation, efficient resource allocation, adherence to the project schedule, stakeholder communication and avoiding any future issues that could occur throughout the project.

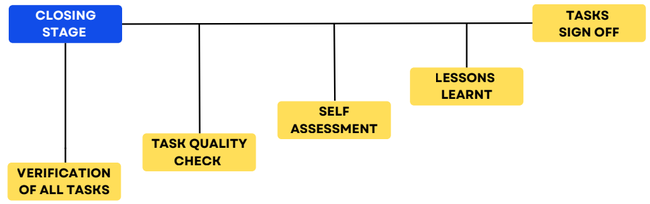


/// *Figure 7.2 | Flowchart showing Action Plan for Scope Creep identification. (Credit: Infinity Racing)*

6.3 /// CLOSING **STAGE**

The Closing Stage included the completion and verification of all tasks and project deliverables. The entire team reflected upon the entire project. This encompassed the Closing Stage where each task was verified and if the quality of tasks was met with the initial Quality Acceptance Criteria for each deliverable.

Moreover, each team member assessed themselves and the lessons learnt through the project were discussed. The Self-Assessment and Lessons Learnt were an important aspect of the Closing Stage as the project came to an end. Lastly, to authorise the completion of the project all tasks were signed-off by the Team Managers to officially declare each task and the entire project as 'complete'.



VIEW OUR CLOSING STAGE PROCESS

SCAN